UNIT-1

Introduction to Design Thinking

Design thinking is a **blend of logic, powerful imagination, systematic reasoning and intuition** to bring to the table the ideas that promise to **solve the problems** of the clients with desirable outcomes. It celebrates the absence of specific formulae, algorithms, or templates for design thinking. A formulaic, simplistic approach can severely limit creative possibilities for solving problems or even finding the right questions. A thorough analysis of the problem's context as well as of the stakeholders themselves, therefore, are critical components of design thinking. Moreover, because design thinking is a process, the ideas are scalable so as to address problems large and small. To keep it simple, Design Thinking is defined as a **flexible** and **systematic** way of **problem solving**.

Design thinking facilitates the understanding and framing of problems, enables creative solutions, and may provide fresh perspectives on our physical and social landscapes. The concept of design thinking began only with a few domains (e.g. architects or product developers etc.), under consideration, but is now found to be applicable to a myriad of disciplines, to solve real-world problems and reconcile dilemmas ranging from medicine and aeronautics to management, operations, and human resource planning.

Origin of Design Thinking

The term "design thinking" has been applied to two different but related areas of conceptual development. The first established body of knowledge, starting in the **late 1960's**, investigates how designers (eg: architects, industrial designers, graphic artists) perform their craft and seeks to identify the skills, abilities and knowledge of expert designers. The second use of "design thinking" is related to business management. Since the mid-**1980's**, interest within the business community grew in exploring how "designerly" thinking could be applied to business challenges, and be performed by employees and leaders not trained formally as designers.

For most practitioners, the idea of design as a way of thinking and solving complex problems in a simplified manner in sciences, can be traced backed to **Herbert Simon** and his **1969** book, ***'The Sciences of the Artificial'***. An American political scientist, economist, sociologist, psychologist, and professor at Carnegie Mellon University, his distinction between critical thinking as an analytic process of "breaking down" ideas and a design-centric mode of thinking as a process of "building up" ideas is foundational to the practice. The same purpose was achieved for design engineering by the book ***'Experiences in Visual Thinking'***, authored by **Robert McKim** in **1973**.

In **1987**, **Peter Rowe**'s book titled, ***'Design Thinking'*** described methods and approaches that planners, designers, and architects use. The work of Robert McKim was consolidated by Rolf Faste at Stanford University during 1980's to 1990's and then, David M. Kelly adapted design thinking for business interests. **David M. Kelly** founded **IDEO** in **1991.**

The term 'design' which has traditionally been associated with a product's aesthetics and visual expression and, in industries where the products 'consist of design' – such as fashion, interior and architectural design - became increasingly common to relate design to the ability to solve complex issues, also commonly referred to as "wicked problems". In **1992**, design theorist **Richard Buchanan** connected fellow design theorist Horst Rittel's term "*wicked problems*" to Simon's Design Thinking when he published **"*Wicked Problems in Design Thinking.*"** In this landmark essay, Buchanan proposed using design to solve the world's most extraordinarily persistent and difficult challenges.

Design Thinking first gained widespread popularity outside the design discourse when it entered the management discourse. This is mainly credited to the work of **Tim Brown** (Brown, **2008**), the former CEO of IDEO, one of the most well-known design consultant firms in the US and Roger Martin, former Dean of the Rotterdam Business School. In one of the most known articles about design thinking, Brown presents design thinking as a human-centred approach to innovation inspired designer's way of working and thinking, which people with non-design backgrounds can use (**2009**).

The Design Ladder

The expanding uses of design have resulted in a need for a way to categorize the different types of design applications. The Design Ladder, developed by **Danish Design Centre** in **2001**, is a scale developed to measure the level of design activities in businesses. Quite a few papers have been written about it since.

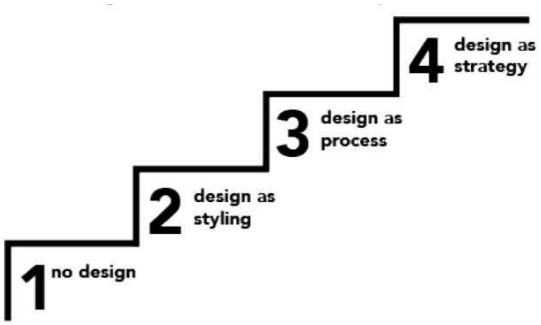


Figure 1.1: The Design Ladder that measure how the design is used in organizations. Recreated from (("Design Ladder," n.d.))

The four rungs of the Danish Design Ladder:

RUNG 1: NON-DESIGN - Design is **invisible**. Product development or marketing initiatives are undertaken by **untrained designers**. Things are produced by the ideas of a small number of people. The user or customer perspective does not really play a part in decisions.

RUNG 2: DESIGN AS STYLING - Design is looked at as a way of making something look nice at the end of the process. It's about **aesthetic**. For example, a product is developed and then given to a product designer or graphic designer to make look nice at the end. Little thought is given to the design of the overall experience.

RUNG 3 DESIGN AS PROCESS - Design is not considered as a result but as a **way of thinking**. Design methods are inbuilt into the early stage of product or service development. Solutions are driven by customer-centric problems and collaboration from multiple teams is employed to develop solutions and communications. The whole process is customer-centric.

RUNG 4: DESIGN AS STRATEGY - Design at this level is **embedded in the leadership** team. Design is used to help in shaping the overall business concept. It is employed to create a vision of the company for the future and then to forge the ways the company is going to get there.

The higher one is on the design ladder the more earnings one will obtain - this was the conclusion of a research paper based on the Danish Design Ladder published in 2003 by Anders Kretzschmar.

In an article by Bryan Hoedemaeckers in 2016, a Director at Deloitte in Australia, he advocates two additional steps to the ladder: "Systemic Change" and "Culture". Matt Davies has taken the liberty of adapting the extended ladder by adding two additional rungs in an article (2019):

RUNG 5: DESIGN AS CHANGE - At this rung, an organisation is using design to **implement its strategy**. It uses tools like roadmaps and diagrams to show where the business is today and where it needs to go and the steps leading to that change. It understands and appreciates things like "change management", "leadership", and "storytelling" in getting its strategy to stick. Muli-disciplined focus groups are used to check-in and monitor change progress. Each phase of change is considered and "designed".

RUNG 6 - DESIGN AS CULTURE - Design is used to create, build and harness a great culture. Design is used to **create desired behaviours** in your people leveraging culture activities, training, learning and development, leadership and employee experience to get the whole organisation aligned around your core purpose. Design is embedded at every stage and there is space for innovation, ideas and leads through customer-centric brand thinking.

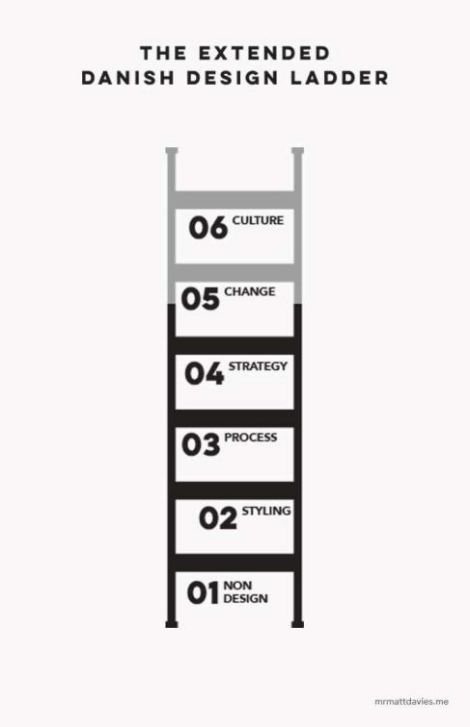


Figure 1.2: The Extended Design Ladder (Adapted by Matt Davies)

Carlgren (**2013**) points out that some might object to leaving out design aesthetics because it can be a driver for innovation, as well as design as processes and strategy. The design ladder model is useful for understanding the distinctions between the different conceptions of design and showing how design thinking is related to design.

Design thinking has been discussed in the **design discourse** for the **last decade** (Hassi & Laakso, **2011**). One of the most notable works on design thinking that tries to clarify the confusion and differentiate the many concepts around design thinking is that of Johansson –Skoldberg et al (**2013**). They define two concepts: "designerly thinking" and "design thinking". **Designerly thinking** belongs to the **design discourse**, describing the practice and reflections of professional designer. **Design thinking**, on the other hand, refers the design practices and competence that include uses **beyond design context** and used by and for people with **nonprofessional design backgrounds**.

Importance of Design Thinking

Design thinking is beneficial for designers to find **innovative, desirable and never-thought-before solutions** for customers and clients.

What is so great about thinking like a designer? Firstly, it is a **tried and trusted problem-solving thinking tool** that anyone can employ to achieve real, concrete results. However, if one did not use it consistently enough, and actively integrate it in their mind-set, one may not be able to see effective results.

The teaching and acquisition of design thinking skills has assumed so much importance that it is now being **taught** at some of the leading universities of the world, as well as the leading global corporate houses across the globe. **Stanford University** in the United States and the **University of Potsdam** in Germany have also promoted design thinking, citing it as one of the most useful skills for professionals.

Infosys Ltd., India's second largest IT-based company providing business consulting, information technology and software engineering services, has also made design thinking a mandatory skill to be acquired by each of its employee.

Respected design critic and educator **Don Norman** suggests that one of the major problems related to innovation is the **ability to manage the desirable, feasible, and economically viable**. Although he is referring to designing things, the problem is equally present when designing systems, services, or even cultures.

3 key factors why people need design thinking:

*1 — **For companies to innovate***

"Innovation distinguishes between a leader and a follower." —Steve Jobs, Apple CEO

Companies need to innovate or die. Today's consumers are very aware, are spoiled for choice, and can be very fickle.

3M and **Apple** were companies on the brink of failure—but they proceed to adopt an innovative culture through design thinking to achieve amazing records of success. Today, 3M generated nearly $30 billion revenue selling over 55,000 innovative products, while Apple captivated the consumer electronics market with stylish products like the iPhone, MacBook and iPad.

In a competitive market, companies, big or small, need to **innovate to create an advantage over their competitors**. Design thinking will allow companies and start-ups to innovate and **explore opportunities**, based on **unmet consumer needs** and understanding of the situation.

2 — For society to solve human problems

"Design is directed toward human beings. To design is to solve human problems by identifying them and executing the best solution." —Ivan Chermayeff, Designer & Artist

People need design thinking to solve human problems (**difficulties we encounter in daily life**). For example, we face issues like overcrowding at train stations, bad online banking experiences and traffic jams, which can lead to frustration and social problems. By applying design thinking principles, we **identify the best possible options to streamline** and make these **experiences go better, faster and smoother**.

3 — For individuals to compete

"Design thinking is a human-centered approach to innovation that draws from the designer's toolkit to integrate the needs of people, the possibilities of technology, and the requirements for business success." —Tim Brown, President & CEO of IDEO

Design thinking has already been seen as a competitive advantage for individuals who want **to excel in their career or business**. Many schools, such as the Rotman School of Design and Stanford University's d.school, have offered **exclusive courses on Design Thinking to executives and professionals**. They aim to give students with design backgrounds a **stronger business advantage**, while offering business-minded people a more **creative edge**. According to the Wall Street Journal, prospects are willing to pay up to $40,000 per year for a Design Thinking MBA, while a Stanford's three-day Design Thinking boot camp costs $9,500!

Thinking like a designer can **transform the way organizations develop products and services on the front end**. It can also **improve processes and strategy in the backend**. You can apply the principles to a new product or service, but you can also use to tackle a problem that a plagues a city or nation.

Understanding Design Thinking- a nonlinear Process

The **iterative design process** helps the designers to **involve clients and customers in meaningful ways**. It is not just a strategy to come up with feasible solutions to a problem, but also a method to think of unimaginable solutions and then trying to make them not just feasible, but also viable.

Design thinking is a solution seeking process that incorporates:

- **User-centred** or human-centred design that commonly refers to a design process that seeks to understand the user as a person as well as incorporate the user's objectives, needs and pains in design efforts. The user is

consulted throughout the design and implementation stages of a solution in order to seek their insights and feedback on how the solution can be made better for the user.

- **Iteration** that repeats cycles of activities that seek to identify and achieve a **viable solution**.
    - One cycle consists of three activities: **learning, making, and evaluating**
    - One cycle consists of two types of thinking: **divergent and convergent thinking**
- **Multiple candidate solutions** instead of initially committing to one "nearly correct" solution early in solution development.
- **Low cost and ideally quick mock ups or prototypes** of solutions for evaluation within the design team and by users
- Direct and collective **participation, observation and evaluation** by members of the design team. These qualities of design thinking are the essence of professional design practice applied to business cases

Consistent with the spirit of design thinking oriented problem solving, multiple systemized approaches to design thinking have been proposed to solve the problem of implementing design thinking in practice. The **Hasso Plattner Institute** uses **six steps** while there is a **four-step** approach from the **Kanazawa Technical College** in Japan. The design consulting company IDEO has yet a different approach. A collection of design thinking approaches can be found at

https://designthinkingmethodology.weebly.com/methodologies.html

Double Diamond Process

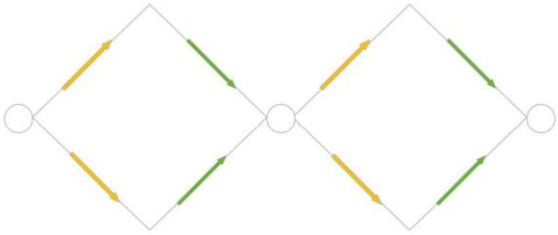


Figure 1.4: Double Diamond process model

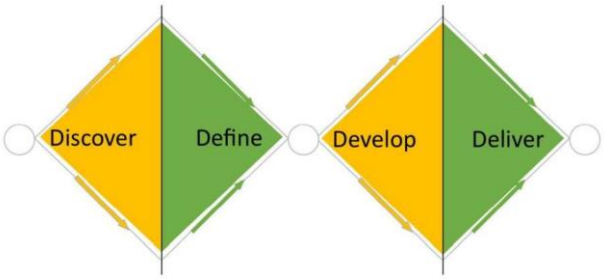


Figure 1.5: Double Diamond process stages

The Design Council, a United Kingdom charity with focus on design, was founded in 1944 to understand and promote design in order to improve businesses within the United Kingdom. Through their research they developed the Double Diamond process model. The name comes from the visual that depicts the process model. In Figure 1.4, each diamond is initiated with divergent thinking followed by convergent thinking.

There are four stages in this model. As shown in Figure 1.5, the stages are Discover, Define, Develop and Deliver. Iteration is recognized as a necessary dynamic in good design, but this process flows primarily left to right. Iteration is expected to occur within the stage. As shown in Figure 1.6, there are three primary artifacts in this process. The first is an initial description of the problem or hypothesis to provide focus to the Discover stage. The second is the formal definition of the problem that results from the Define stage, which is captured in a document called the design brief. The design brief establishes the foundation for solution development performed in the Develop stage. The Deliver stage produces a solution resulting from a final design, which is produced and introduced to customers.

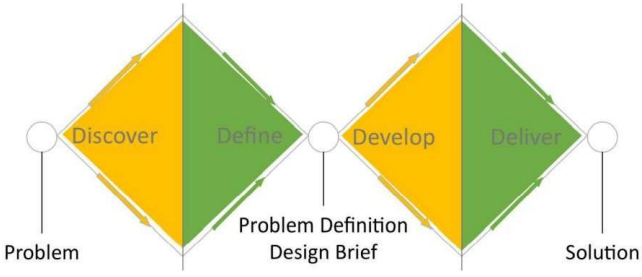


Figure 1.6: Double Diamond primary artifacts

In the **Discover stage**, design thinkers expand on the initial problem description by using design methods that expand understanding, get different points of view, and notice things about the problem they had not earlier considered. The **Define stage** is where the design thinking team builds a coherent understanding informed by considering the insights obtained in the Discover stage. The team prioritizes aspects of the problem in terms what should be acted upon first. They consider feasibility as a criterion. The goal is to develop the design brief that will be the basis of upcoming development efforts. The next stage, **Develop**, is dedicated to the creation, testing and refinement of solution concepts and prototypes. The final stage, **Deliver**, signifies the selection and final adjustments of the solution. The solution is manufactured or implemented and made accessible to the targeted population.

Designing for Growth Process

The "Designing for Growth" (or Growth) process model was introduced by **Jeanne Liedtka** and **Tim Ogilvie** in a book published in **2011**. At the time of writing the book, Jeanne Liedtka was a member of the University of Virginia's Darden School of Business. Tim Ogilvie was the CEO of an innovation strategy consultancy called Peer Insight. The book has a related website with information about this book and its companion book.

The Growth process model consists of four stages that are framed as questions: 'What is?', 'What if?', 'What wows?', 'What works?' The general flow through the process model aligns with the order of the questions. The authors describe ten tools that align with their process model. They also recommend the use of four project management aids. The visual associated with the Growth process model is unique and is similar to the Double Diamond in that it aligns the use of divergent and convergent thinking to the process. The original depiction of the Growth visual can be found at the website mentioned in the previous paragraph. Figure 1.7 shows an interpretation of their visual. The intensity of colored fill indicates the intensity of divergent and convergent thinking occurring within each stage. The What if stage is the most intense in terms of divergent and convergent thinking followed by the 'What wows' stage, 'What is' stage and finally the 'What works' stage.

With respects to iteration and the expected flow through the Growth process, the expectation is that an actual project utilizing the Growth process model will not flow smoothly from left to right and that iterative design is critical to closing in on a viable solution. Liedtka and Ogilvie caution that these tools may align slightly differently depending on the needs and opportunities within your project.

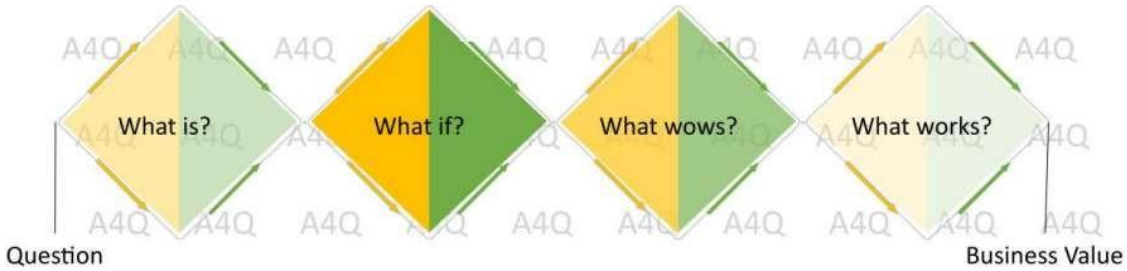


Figure 1.7: Growth process model – an interpretation

For purposes of comparison, a brief description of each stage is provided below:

– **What is?** - The design thinking team in this stage investigates and **seeks to understand the users** and their current needs, wants and pains related to the problem.

– **What if?** - In this stage, the team imagines and conceives possible future solutions.

– **What wows?** - In this stage, the team selects and focuses on concepts that wow, which requires testing assumptions and prototyping.

– What works? - In this stage, users have an opportunity to experience the prototype and make creative suggestions. The most promising prototype is transformed into a nearly final product/service and rolled out to a test market.

5-Stage Stanford d.School Process

The Stanford d.School, more formerly called the Hasso Plattner Institute of Design at Stanford, is an academic collaboration between the Hasso Plattner Institute in Potsdam, Germany and Stanford University in Stanford, California. The Stanford d.School was one of the first d.Schools or Design Schools formed around design thinking approaches to design. Their process model has changed from a 6-stage model of Understand, Observe, Point of View, Ideate, Prototype, and Test to a 5- stage process model of Empathize, Define, Ideate, Prototype and Test. The stages of Understand and Observe were consolidated to Empathize according to Lewrick et. al. The stage titled Point of View was adjusted to become Define in the current model.

Unlike the Double Diamond the visuals associated with the 5-stage process model are not firmly set. In Figure 1.8, the author's interpretation of possible stage dynamics is depicted. Iterative transitioning between stages is expected. After initial testing has started, iteration between Test, Define and Prototype (possibly Ideate as well) is expected.

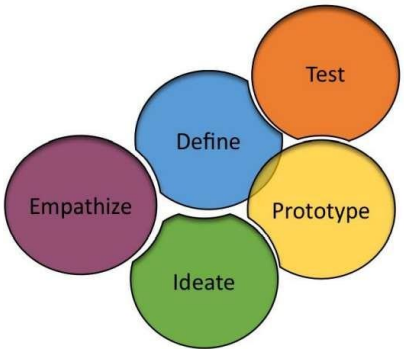


Figure 1.8: Stanford d.School 5-Stage process model stylized to show a possible stage dynamic

Beyond suggesting the 5-stage process model, Stanford d.School advocates for six attributes of the design thinking mind-set: human centred, bias toward action, radical collaboration, culture of prototyping, show don't tell, and mindful of process. Human centred is the idea that design thinkers should identify with the user's challenges and develop solutions that address their needs. Bias toward action directs design thinkers to solve problems instead of finding additional ones. Team members should prototype to learn insights instead of expressing self-important personal opinion. Radical collaboration suggests breakthroughs come from a team consisting of members with diverse viewpoints, education and experiences. Culture of prototyping is related to the notion that solutions will arise from trying out concepts and getting user input early and often. "Show don't tell," emphasizes the need to communicate visually rather than verbally. Mindful of process encourages teams to be aware of which stage the project is in, and what needs to be accomplished (ex. divergent thinking during Ideate but convergent thinking during Define) in the current stage.

In the Stanford d.School's process model, the stages have the following objectives:
In **Empathize stage**, the design thinker **puts himself or herself into the shoes of the end user** and tries to understand the needs of the customer and the problem from their viewpoint by observation, engagement and immersion. A lot of interview, field visits, etc. are required for gathering the information on the requirements. This phase has the customer directly involved in the design thinking process.

Once the requirements are clear, the **Define stage** helps to frame an **actionable problem statement** or **design vision** tailored for the users.

In the **Ideate phase**, within the context of the problem statement, a design thinking team **generates many "radical" design alternatives** that explore the solution space. The influx of ideas is represented as a mind map or as a storyboard or a document.

In the **Prototype phase**, a design thinker focuses on testing the ideas on the grounds of **feasibility and viability**. The unfeasible ideas are discarded and the feasible ones are converted to prototypes. Promising design alternatives are made **tangible** with which the team, users and others will experience and interact. The process of prototyping helps the design thinker to **understand the issues** related to an idea, which were never thought before. This helps the team of design thinkers to come up with the best prototype and decide the best solution at hand. Moreover, the customer is directly involved in this phase and its feedback is critical for the design thinkers.

In the **Test phase**, the prototype or the model is presented to the customer with the goal of gathering thoughtful **feedback**, learning and **refining solutions**. The feedback from the end user decides whether the solution suggested by the design thinkers has been fruitful or not. If the end user does not approve of the solution, then the entire process has to be iterated. The concept of **iteration** is hence central to the process of design thinking.

Design thinking not only helps to come up with innovative solutions, but also helps to address the exact problems faced by the customer and target the customer's requirements in the best possible manner.

Design Action Plan

The design action plan is a framework that contains a series of action phases that execute the design thinking process. It is a roadmap that tells people involved in a project whether they are going to be on the right track.

5 CHARACTERISTICS OF ACTION PLAN

1 — It is not just a brainstorming session or a "one-day" process.

The design action plan is not a process that works instantaneously - it requires time and effort from committed individuals involved with the same objective to make it work. It is certainly not a "one-day" process where problems can be solved in 24 hours. There are phases where you need multiple iterations, testing and checking back with users and exploration where it would take time to incubate. Some have commercialized design thinking as a "brainstorming" session. It is not. Remember, design thinking is a 5-step process, and not a 1-step Ideate Phase. It seeks to define the problem, even as it finds the solution.

2 — It is an iterative process.

The design action plan is an iterative process. You need to repeat each phase backwards and forwards, and arrives at each decision or desired result, after rounds of discovery. It means to work on something over and over again, until every facet becomes crystal clear.

For example, you can iterate between Ideate and Prototype Phase several times - you sketch ideas and immediately build prototypes to make rounds and rounds of discovery of the idea. After that, you may want to go back to Ideate Phase to refine the ideas yet again.

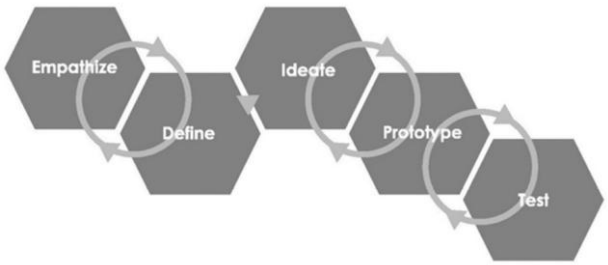


Figure 1.9 Iterative Process of the five phases

3 — It contains phases of both divergent and convergent thinking.

The design action plan has phases of both divergent and convergent thinking.

The Empathise and Ideate Phase requires divergent thinking. Divergent thinking is to create and explore ideas and possibilities, without restraint. Within the phase, you will be encouraged to make guesses, come out with wild ideas and be creative.
The Define, Prototype and Test Phase requires convergent thinking. Convergent thinking means to actively select and decide the right solutions to the problem, based on constraints or feedback. Within the phase, you will be encouraged to analyze and make decisions and constantly ask critical questions to solve the problems at hand.
Both sides of thinking are important in order to make this process work.
"Divergent thinking is to create and explore ideas and possibilities, without restraint."
"Convergent thinking is actively selecting and deciding the right solutions to the problem based on set of constraints or feedback."

4 — It requires everyone to collaborate and go through the process together.

The design action plan is made up of phases where everyone, including the design thinking facilitator, collaborates on the process together. Everyone will bring their prior experiences and knowledge about the problem to the table and work on the issue together.
The process is not like a hot potato, where you "toss" the responsibility from one individual to another. For example, the researcher in the Empathize Phase might share to the business lead in the Define Phase and define the project, which engages a designer to coordinate the Ideate Phase, and then leave everything to the engineer in the Prototype and Test Phase!
But this is not proper design thinking. Instead, all of them (the researcher, business lead, designer and engineer) should be in the process from start to end.
This is because everyone in the team requires deep insight of what the issues are and what are needs of the users - so that everybody is on the same page. Design thinking is a process where everyone collaborates together from start till end with a common objective.

5 — There are deliverables at every phase.

There is a deliverable outcome of each action phase which will give you a point of reference, when you go through the process of iteration. This list of deliverables is not exhaustive, but you do not have to use them all. Based on the nature of your project, select the necessary deliverable.

At the end of each phase we have:



Figure 1.10 Deliverables at the five phases

Table 1.2 Characteristics of Action Plan - Summary

Characteristics	What would you expect	What is expected of you?
1. Not just a brainstorming session	Proactive process of action phases over a period of time	Be patient and prepared that the process does not work like instantaneously
2. Iterative process	Repeated rounds of discovery of ideas / concepts until you become crystal clear	Be willing to discover and never stop at the first idea / concept
3. Has phases of both divergent and convergent thinking	Explore ideas & possibilities at one action phase while being decisive in solutions in another	Approach different way of thinking at each action phase
4. Requires collaboration	All stakeholders go through the process together	Be open and accept others into the process- everyone has experience and expertise to bring to the table
5. Deliverables at every phase	Point of reference at every action phase	Advocate to document deliverables at every phase

Applications of Design Thinking

Design thinking is used extensively in the area of healthcare and wellness, agriculture, environmental sustainability, food security, sports, education and research to business, management, to name a few. Design thinking has helped in the digital space, contributed to the development of physical products, spurred social innovation projects and much more. Design thinking can be applied to community settings (ex. economic revitalization, social services) and education (ex. student learning, learning experiences).

An example would be to take a gadget and ask, "How do I make this work, better, faster, smoother, and reduce waste and inefficiency?" But it can go beyond product design. You can scale it up and apply it to many things.

Design thinking is a way to get businesspeople to think like designers and designers to think like businesspeople. But design thinking is more than that.

The basic principle of design thinking is that innovation can be disciplined. Innovation is not an elusive entity that only a few genius people can experience. It is, rather, a practice that can be systematically approached by a set of practical and meticulous tools, methodologies, and frameworks.
Design thinking helps you learn the following:

- How to optimize the ability to innovate?
- How to develop a variety of concepts, products, services, processes, etc. for end users?
- How to leverage the diverse ideas of innovation?
- How to convert useful data, individual insights and vague ideas into feasible reality?
- How to connect with the customers and end-users by targeting their actual requirements?
- How to use the different tools used by designers in their profession for solving your customers' problems?

Design thinking helps in Optimization of capabilities, Feasibility and viability analysis, and Addressing the needs of customers. Design thinking helps people of every profession to arrive at solutions in a planned, organized, and systematic manner.

Business uses of Design Thinking

Design thinking helps in businesses by optimizing the process of product creation, marketing, and renewal of contracts. All these processes require a companywide focus on the customer and hence, design thinking helps in these processes immensely. Design thinking helps the design thinkers to **develop deep empathy for their**

customers and to create solutions that match their needs exactly. The solutions are not delivered just for the sake of technology.

The value that design thinking has brought to organizations is **innovation** that has disrupted their markets, enhanced brands in terms of recognition, sales and performance, and business adaptation. This value has been achieved by innovating in the areas of product development (ex. Proctor & Gamble's Swiffer), service development (ex. AARP's LifeTuner.org), business models, and business strategy.

Design thinking - Education Sector

One domain in which design thinking finds a huge application is the education sector. These days, educators are using design thinking extensively to improve the quality of education in schools, especially in the kindergarten classes. Design thinking has been used in schools to **upgrade the curriculum**, or to **redesign the student spaces** or to make the entire experience of the students worthwhile.

Design thinking helps the school administrators to solve institution-based problems, helps in making the curriculum more valuable to the students and to stimulate design thinking skills in students as well.

By taking feedback from students on their requirements, goals and challenges they are facing in the classroom. By working on their feedback, the instructors can come up with solutions to address their challenges.

For example, **Michael Schurr**, a 2nd grade instructor from New York, realized that his students would be more comfortable with bulletin boards lowered. He also found the idea of creating comfortable semi-private space for working students as it provided them space to study. As a result, his students became more engaged and felt free to move.

REDLab Group

Stanford University's Graduate School of Education has a REDLab group which conducts research on how to apply design thinking in kindergarten, 1st to 12th grade, secondary and post-secondary academic settings. The REDLab group has teamed up with Hasso Plattner Institute to create Hasso Plattner Design Thinking Research Program, which works for applying rigorous academic methods to understand the reasons behind the success and failure of design thinking programs.

Tools at Schools

The Tools at Schools group once conducted an activity with the 8th grade students in The School at Columbia University. The activity included redesigning a locker, chair or desk to suit the requirements of 21st century students. Design thinking was applied in this activity and the results were displayed in International Contemporary Furniture Fair. Design thinking also helps the pupils to learn how to work collaboratively in a team.

https://www.oliviergregoire.com/Tools-at-schools

Design Thinking - IT Industry

The IT industry makes a lot of products that require trials and proof of concepts. The industry needs to empathize with its users and not simply deploy technologies. IT is not only about technology or products, but also its processes. The developers, analysts, consultants, and managers have to brainstorm on possible ideas for solving the problems of the clients. This is where design thinking helps a lot.

The IT industry is now working mostly on Agile methodology, which is a technique of project management.

Agile is an iterative or incremental method of managing development and design. Each day, the engineers have a set of tasks at hand, which they are required to complete in a day or two. Moreover, the complex problems shot by the customers to the engineers are required to be solved quickly. In such a scenario, design thinking helps to solve the problems and address the exact needs of the customer.

In a Waterfall model, which is different than the Agile model, the process starts with requirements gathering, followed by creating the visual designs and then occurs the development of solution. Testing is the last step in the model. Looking closely, it is similar to the process of design thinking.

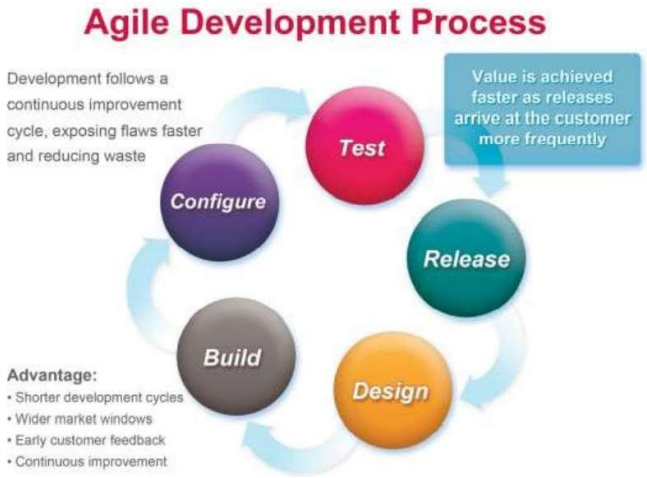


Figure 1.11 Agile Methodology

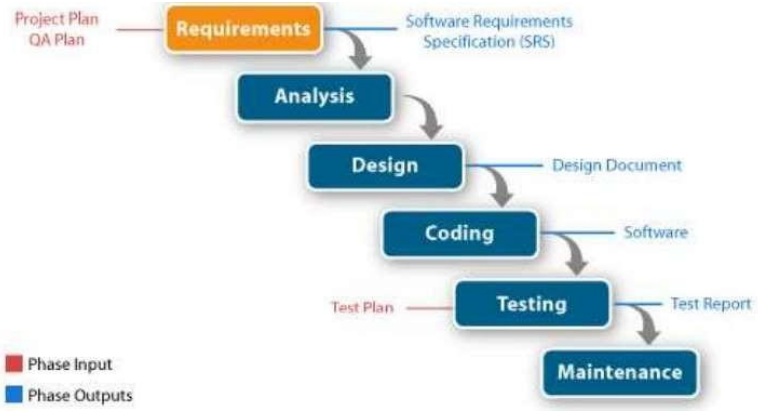


Figure 1.12 Design Thinking in IT industry

Design Thinking - Healthcare Industry

Design thinking can certainly help the healthcare experts to come up with solutions that plague the healthcare system. The basic question for every healthcare expert is 'how to provide quality healthcare at low and affordable prices to everyone?'

Venice Family Clinic in Venice, California has come up with innovative solutions to the challenge of opening a low-cost children's clinic to serve the low-income families. Problems of finance, transportation, and language barriers had to be solved. And all this had to be done at low cost for the poor kids. Fostering good health along with profits was a challenge, as it does not sound sustainable. Using design thinking, the inefficiencies in the system and the perennial crises were addressed. This was followed by mind-blowing innovations to serve the children.

https://www.careinnovations.org/resources/using-design-thinking-to-prescribe-food-as-medicine/

Design Thinking - Social Innovation

Social issues are always complex problems, which have too many strands attached to them. There are too many aspects of a problem, that many a times get ignored by the social innovators. However, solving a social problem requires taking into consideration all the facts and figures, and then working on them. This is the reason why

design thinking is being widely used for social innovation. As a result, non-profits have begun to use design thinking extensively these days.

IDEO Example

In 2008, Bill and Melinda Gates Foundation asked IDEO to codify the process of design thinking. The foundation wanted the code to be used by grassroots level NGOs to solve problems for small farmers in the developing nations. A team from IDEO worked for months in association with the International Centre for Research on Women, Heifer International, and International Development Enterprise to get insights into the process of designing new products.

https://designthinking.ideo.com/blog/human-centered-design-for-small-holder-farmers

Naandi Foundation's Example

In the city of Hyderabad in India, Naandi Foundation's community water treatment plant provides safe water. However, villagers still use free water which is not safe for consumption and makes people sick. The villagers use unsafe water not because of affordability issues or accessibility issues, but because of the flaws in the overall design of the system.

The problem is that the womenfolk cannot bring the heavy containers of water back to their homes from the plant. Such problems can be solved by design thinking process. A trolley-like structure is made to carry and drag the heavy water cans easily on wheels. A simple yet impactful solution.